A **prompt** is the **instruction or input text given to an AI model (like ChatGPT or other LLMs)** to guide what kind of response it should generate.
It tells the model **what task to perform, what context to consider, and how to format the answer**.

Example prompt:

"Explain the concept of photosynthesis in simple terms for a 10-year-old student."

Here the AI understands:

- Topic → Photosynthesis
- Style → Simple explanation
- Audience → 10-year-old

**Breakdown of the 4 Parts:**

- **Role/Persona:** Defines the perspective or expertise the AI should adopt (e.g., "Act as a senior marketer").
- **Context/Background:** Provides necessary details to set the scene (e.g., "We are launching a new eco-friendly water bottle").
- **Instruction/Task:** Specifies the exact action required (e.g., "Write a 50-word product description").
- **Output Format:** Determines how the result is presented (e.g., "Use bullet points," "Format as a table").

**Types of Prompts in LLM Systems**

In chat-based AI systems there are typically **three prompt types**.

1. System Prompt
2. User Prompt
3. Assistant Prompt

**System Prompt**

A **system prompt** defines the **overall behavior and rules of the AI**.

It is usually written by the **application developer**, not the end user.

Example:

You are an expert legal assistant.
Provide accurate answers based on Indian law.
Always explain in a professional tone.

**Purpose**

- Define the AI's role

- Control personality and tone
- Set safety rules
- Guide overall response behavior

**User Prompt**

A **user prompt** is the **question or request asked by the user**.

Example:

What are the fundamental rights in the Indian Constitution?

**Purpose**

- Specifies the user's request
- Determines the task the AI should perform

**Assistant Prompt**

This is the **AI's previous response** in the conversation history.

Example:

Assistant: Fundamental rights are basic human rights guaranteed by the Constitution...

It helps the AI **maintain conversation context**.

| Feature | System Prompt | User Prompt |
|---|---|---|
| Who writes it | Developer / application | End user |
| Purpose | Define AI behavior | Ask a question or give task |
| Priority | Higher priority | Lower priority |
| Visibility | Usually hidden | Visible |
| Scope | Global rules | Specific request |
| Example | "You are a legal expert." | "Explain contract law." |

Example Showing Both

**System Prompt**

You are a medical assistant. Give safe and general health information only

**User Prompt**

What are the symptoms of dengue?

**Assistant Output**

Common symptoms of dengue include:
- High fever
- Severe headache
- Joint and muscle pain
- Rash

## Example of a Complete Prompt

Role:

You are a senior marketing expert.

Context:

A company is launching a new eco-friendly water bottle made from recycled materials. The target customers are environmentally conscious young adults.

Task:

Write a short product description promoting the eco-friendly water bottle.

Format:

Write the description in 50 words using bullet points.

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Role:

You are an experienced history teacher.

Context:

Students are preparing for their school exam on World War II.

Task:

Explain the main causes of World War II.

Format:

Provide the answer in 5 bullet points using simple language.